

Siddarth M -Python Developer

📍 Chennai ✉ siddarthmofficial@gmail.com ☎ 6379399560 📄 in/siddarth-m-43ba61246 🌐 siddarth-py.github.io/

SUMMARY

Experienced Software Engineer with **5+ years of software development** experience and strong expertise as a **Full Stack Developer**. Proficient in **Python and FastAPI**, with hands-on experience building RESTful APIs, optimizing databases, and developing scalable, high-performance applications. Skilled in creating responsive frontend applications and collaborating within Agile/Scrum environments to deliver clean, maintainable, and efficient code. Currently expanding expertise in **Artificial Intelligence (AI) and Generative AI (LLMs)**, including the **Segment Anything Model (SAM)**, to build intelligent systems and data-driven analytics solutions.

EDUCATION

B.E-Computer Science
K.Ramakrishnan College Of Technology•Trichy, Tamilnadu•2014-2018

SKILLS

Programming Languages:Python, C, Embedded C, JavaScript.
Backend Frameworks:Django, FastAPI, Flask, Rest Framework.
Frontend Technologies:HTML, CSS, Bootstrap.
AI / Computer Vision:PyTorch, Segment Anything Model (SAM), OpenCV,Matplotlib.
Databases:MySQL, PostgreSQL, SQLite.
Tools & Platforms:Postman, Docker, Git, GitHub, JIRA, IBM DOORS, Enterprise Architect, PTC.

EXPERIENCE

PythonfullStackDeveloper

Mileveen Technologies **December 2025 - Present,Chennai**
•Designed and deployed scalable **RESTful APIs using FastAPI**, improving modularity and maintainability of backend services.
•Implemented optimized **CRUD workflows**, reducing response latency and improving transaction reliability.
•Applied **Clean Architecture principles**, enhancing code reusability and reducing technical debt.
•Integrated **Pydantic validation models**, improving data integrity and minimizing invalid API requests.
•Contributed to **AI-driven** backend modules supporting **computer vision-based retail analytics solutions**.

SoftwareEngineer

Ielektron Technologies **July 2022 - December 2024,Chennai**
•Implemented **JWT-based authentication and RBAC**, strengthening API security and access control mechanisms.
•Developed full-stack applications with **Django REST Framework** and responsive frontend interfaces.
•Designed and optimized relational database schemas, improving query performance and data consistency.
•Integrated **backend APIs** with frontend systems for seamless real-time data interaction.
•Utilized **NumPy and Pandas** for backend data processing and structured analysis workflows.

SoftwareEngineer

Webberax **November 2018 - May 2022,Chennai**
•Developed and maintained **SYS2 and SWE1** requirements aligned with automotive standards and ASPICE processes.
•Authored detailed Functional Test Cases, improving validation coverage and system reliability.
•Designed embedded control features for automotive seat systems using **LIN and CAN protocols**.
•Implemented **Seat Memory and Welcome Recall features ensuring seamless ECU communication**.
•Integrated **ISO 26262** functional safety standards, enhancing system robustness and audit compliance.

PROJECTS

Retail_Shelf_Analyzer

Role: AI / Backend Developer - Mileveen Technologies

•Designed and developed an **AI-based computer vision system** to analyze retail **shelf images** and extract structured inventory insights.
•Integrated **Segment Anything Model (SAM) with OpenCV** for automated product segmentation and object detection.
•Leveraged **NumPy and Pandas** for efficient image array manipulation, stock computation, and reporting workflows.
•Implemented **automated stock count estimation and empty shelf detection logic**.
•Built an interactive **analytics dashboard** for real-time shelf monitoring.

Key Achievements:

- Reduced manual shelf audit effort by approximately 40% through automated detection.
 - Accelerated inventory reporting time by automating stock estimation workflows.
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FastAPI_Booking_Management_System**Role: Full Stack Developer -Mileveen Technologies**

- Designed and developed a scalable RESTful **Booking Management API** using **FastAPI** for generic service operations.
- Implemented structured **CRUD** workflows with standardized error handling and validation.
- Applied Clean Architecture principles, ensuring **separation of concerns across routers, use cases, and domain models** for maintainability and scalability.
- Integrated **Pydantic** validation to enhance request integrity and minimize runtime errors.

Key Achievements:

- Improved **API** reliability by reducing invalid request failures through validation mechanisms.
 - Enhanced backend maintainability through **modular architecture implementation**.
 - Delivered a scalable **booking system** capable of supporting multi-service integration.
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Universal Communication Gateway (UCG)**Role: Full Stack Developer -lelektron Technologies**

- Developed a Universal Communication Gateway (UCG) to enable **interoperability across 2G, 3G, 4G, and 5G telecom networks** through protocol translation and standardized communication interfaces.
- Designed and developed **RESTful APIs** for gateway configuration, protocol translation, monitoring, logging, and interoperability testing.
- Implemented **JWT-based authentication and authorization** to secure API endpoints and protect gateway services.
- Built protocol translation and data serialization modules supporting **JSON, Protocol Buffers**, and Avro formats.
- Developed a responsive **web interface for gateway configuration**, monitoring, and administrative operations across desktop and mobile devices.
- Integrated Swagger/OpenAPI for API documentation and standardized **API testing workflows**.
- Containerized backend services using Docker and implemented **CI/CD pipelines** with Jenkins/GitLab CI to automate build and deployment processes.
- Developed a deployment-ready backend **architecture supporting scalable cloud-native infrastructure**.

Key Achievements:

- Improved interoperability between heterogeneous telecom networks through standardized protocol translation.
 - Enhanced **API security** using JWT-based authentication and secure access control.
 - Streamlined deployment and release management using **Docker and CI/CD automation**.
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Autopark_Park_Assist**Role:Software Engineer -Webberax**

- Contributed to development of **Auto Park systems** enabling fully automatic vehicle parking using guidance technology and camera based vision systems.
- Managed **SYS2 and SWE1** requirement documentation.
- Coordinated cross-functional requirement validation and **Analyzed system-level requirements** and managed release of base software versions.

Key Achievements:

- Improved requirement traceability and review efficiency through structured documentation.
 - Supported successful feature validation aligned with automotive standards.
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TWIN_BLDC_MOTOR**Role: Software Engineer -Embedded -Webberax**

- Developed control software for a **BLDC motor-driven air compressor system**, enabling efficient generation of compressed air.
- Implemented electronic commutation for precise **torque and speed control**.
- Integrated SBC control logic on RH850 microcontroller for fault monitoring and Implemented **CRC validation and secure boot mechanisms**.

Key Achievements:

- Enhanced system safety via **power fault protection mechanisms**.
 - Increased **embedded firmware** robustness using integrity validation features.
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